

SEBASTIAN CORREA-GALLEGO

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Research Interests

Life assembles into order under constraint. A microbial community drawn from the same pool of organisms may converge on a single reproducible state or settle into one of several alternatives depending on the order in which its members arrive, and what decides between these outcomes — whether assembly follows rules or depends on history — remains among the open questions in ecology and evolution. This tension between predictability and contingency reaches beyond any single system: it echoes the broader problem of how autonomous parts come to form integrated wholes, from the structure of a community to the major transitions in the organization of life. I am drawn to these questions, and to the possibility that collective outcomes might be anticipated from measurable properties of the organisms themselves — such as how a cell divides its finite resources between growth and metabolic efficiency. I hope to pursue them with growing quantitative rigor while keeping the living organism, in its real environment, at the center of the inquiry.

Education

Universidad EAFIT, B.Sc. in Biology
GPA: 4.44 / 5.00

2022–2026

Research Experience

Visiting Student Intern

Aug 2025–Jan 2026

[The Evolution of Cells, Symbionts and Organelles Lab](#), Dept. of Biological Sciences, Purdue University

- Contributed to the project *Proteome Allocation Rules in Osmotrophic Eukaryotes*, studying how non-conventional osmotrophic yeasts (*K. marxianus* and *V. polyspora*) reorganize proteome investment under carbon limitation.
- Generated high-resolution growth curves to characterize the physiological response of the study organisms across defined carbon regimes, establishing the growth phenotypes underlying the downstream analysis.
- Contributed to proteomics-based workflows, including sample preparation and integrative analysis linking molecular allocation to cellular physiology.
- Tested and calibrated an automated continuous-culture system (Pioreactor) to support controlled cultivation for the lab's steady-state experiments.

Undergraduate Thesis Researcher

2024–2026

Universidad EAFIT, Advisor: Prof. Nicolás Pinel Peláez, Ph.D.

Thesis: *Cultivable Microbial Community Structure Along a Light Gradient in a Tropical Volcaniclastic Cave* [[permanent link](#)]

- Characterized the cultivable microbial fraction of the Organal San Antonio, a tropical volcaniclastic cave at ~2350 m a.s.l. in Támesis, Antioquia, across a light-defined spatial gradient spanning Entrance, Transition, and Dark sectors.
- Found systematic differences in cultivable abundance and community composition across zonation: the aphotic sector supported markedly reduced densities and a distinct assemblage relative to photic sectors, while the Transition zone harbored a partially unique community of its own.
- Established a first spatially resolved cultivable baseline for an organal-type pseudokarstic cave system, designing a tractable cultivation study in the absence of sequencing infrastructure.

Academic Service

Student Director, Research Group on Microbiology and Astrobiology (SIAB) *Oct 2023–May 2025*
Universidad EAFIT, affiliated with the Research Group on Geosciences and Biodiversity (GEBI)

- Led a student research group on microbiology, astrobiology, and extremophiles; taught basic microbiology techniques to incoming students and coordinated projects on cave and thermal-spring microbial communities.
- Directed an outreach project on the role of microorganisms in the maintenance of Andean forests, producing science-communication material and interpretive displays for the El Globo nature reserve (Támesis).

Conferences and Presentations

Research Proposal Presentation *2024*
2nd Symposium of Biology, Universidad EAFIT

Oral Presentation *2024*
XXIII Encuentro Departamental de Semilleros de Investigación, RedCOLSI, Antioquia, Colombia

Research Fair Presentation *2022*
Feria de Semilleros de Investigación, Universidad EAFIT

Honors and Recognition

Fully Funded Internship, Fundación Fraternidad Medellín — UREP-C Program, Purdue University *2025*

Award Winner, Research Proposal Presentation, 2nd Symposium of Biology, Universidad EAFIT *2024*

Undergraduate Scholarship, Comfama and Fundación Fraternidad Medellín *2022*

Certifications and Training

Responsible Conduct of Research (RCR) Training *Aug 2025*
CITI Program, Credential ID 71597371, valid through Aug 2029

Technician in Agricultural Sciences *2019–2021*
Servicio Nacional de Aprendizaje (SENA), Colombia

Technical Skills

Laboratory and biological methods: microbial cultivation and isolation, continuous-culture operation, growth-curve measurement, environmental sampling and ecological field records, molecular biology and biochemistry techniques, proteomics sample preparation, and microscopy-based observation.

Quantitative and computational tools: R, Python, Linux/bash, $\text{\LaTeX}$, and QGIS, applied to biological data analysis, statistics, and data visualization.

Scientific workflows: scientific writing, research communication, literature synthesis, and figure preparation.

References

Nicolás Pinel Peláez, Ph.D.
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Sergio A. Muñoz-Gómez, Ph.D.
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